Definition 1. The category $\mathsf{sVec}_{\mathbb{R}}$ is the symmetric monoidal category of $\mathbb{Z}_2$-graded real vector spaces and degree-preserving linear maps, with tensor product

$$(X \otimes Y)_a = \bigoplus_{i+j=a} X_i \otimes_{\mathbb{R}} Y_j, \tag{1}$$

unit $\mathbb{R}$ concentrated in degree zero, and the usual associativity and unit constraints. Its symmetry is given by

$$\beta_{X,Y}(x \otimes y) = (-1)^{ij} y \otimes x \qquad (x \in X_i,\ y \in Y_j). \tag{2}$$

Definition 2. A *supercategory* is an $\mathsf{sVec}_{\mathbb{R}}$-enriched category, and a *superfunctor* is an $\mathsf{sVec}_{\mathbb{R}}$-enriched functor. An *even supernatural transformation* $\theta : F \Rightarrow G$ between superfunctors $F, G : \mathcal{C} \to \mathcal{D}$ is a family of even morphisms $\theta_X : F(X) \to G(X)$ satisfying

$$\theta_Y \circ F(f) = G(f) \circ \theta_X \tag{3}$$

for every homogeneous morphism $f : X \to Y$. A superfunctor $F : \mathcal{C} \to \mathcal{D}$ is a *superequivalence* if there is a superfunctor $G : \mathcal{D} \to \mathcal{C}$ and even supernatural isomorphisms $GF \cong \mathrm{Id}_{\mathcal{C}}$ and $FG \cong \mathrm{Id}_{\mathcal{D}}$. The *underlying category* $\mathcal{C}_0$ of a supercategory $\mathcal{C}$ is obtained by retaining all objects and only the even morphisms:

$$\mathcal{C}_0(X, Y) = \mathcal{C}(X, Y)_0. \tag{4}$$

An object of a supercategory $\mathcal{C}$ is *simple* if it is simple in $\mathcal{C}_0$.

Definition 3. For supercategories $\mathcal{C}$ and $\mathcal{D}$, the *product* $\mathcal{C} \boxtimes \mathcal{D}$ is a supercategory with objects (X, Y), morphism spaces

$$(\mathcal{C} \boxtimes \mathcal{D})((X, Y), (X', Y')) = \mathcal{C}(X, X') \otimes_{\mathbb{R}} \mathcal{D}(Y, Y'), \tag{5}$$

and composition

$$(f' \boxtimes g') \circ (f \boxtimes g) = (-1)^{|g||f'|}(f' \circ f) \boxtimes (g' \circ g) \tag{6}$$

for homogeneous morphisms. A *monoidal supercategory* is a supercategory $\mathcal{C}$ equipped with a superfunctor $\otimes : \mathcal{C} \boxtimes \mathcal{C} \to \mathcal{C}$, a unit object $\mathbf{1}$, and even supernatural associativity and unit isomorphisms satisfying the pentagon and triangle axioms. Its underlying category $\mathcal{C}_0$ inherits the monoidal structure of $\mathcal{C}$. For monoidal supercategories $\mathcal{C}$ and $\mathcal{D}$, the product $\mathcal{C} \boxtimes \mathcal{D}$ is a monoidal supercategory with unit $(\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{D}})$ and tensor product

$$\begin{aligned} (X, Y) \otimes (X', Y') &= (X \otimes X', Y \otimes Y'), \\ (f \boxtimes g) \otimes (f' \boxtimes g') &= (-1)^{|g||f'|}(f \otimes f') \boxtimes (g \otimes g') \end{aligned} \tag{7}$$

for homogeneous morphisms. Its associativity and unit constraints are defined componentwise. A *monoidal* Π*-supercategory* is a monoidal supercategory equipped with an object π and an odd isomorphism $\zeta : \pi \xrightarrow{\sim} \mathbf{1}$.

Definition 4. A *monoidal superfunctor* between monoidal supercategories $\mathcal{C}$ and $\mathcal{D}$ is a superfunctor $F : \mathcal{C} \to \mathcal{D}$ equipped with an even supernatural isomorphism $J_{X,Y} : F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y)$ and an even isomorphism $\varphi : \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(\mathbf{1}_{\mathcal{C}})$, satisfying the usual monoidal functor axioms. It is a *monoidal superequivalence* if F is a superequivalence. We write $\mathcal{C} \simeq_{\otimes} \mathcal{D}$ when such a superequivalence exists.

Definition 5. A monoidal Π-supercategory $\mathcal{C}$ is a *fusion* Π*-supercategory* if $\mathcal{C}_0$ is a fusion category over $\mathbb{R}$ in the sense of [San24, Definition 2.9]. An object X of a fusion Π-supercategory $\mathcal{C}$ is *weakly invertible* if

$$X \otimes X^* \cong \mathbf{1}^{\oplus p} \oplus \pi^{\oplus q} \quad \text{in } \mathcal{C}_0 \qquad \text{for some } p, q \geq 0, \tag{8}$$

where X^* is a left dual of X in $\mathcal{C}_0$. A fusion Π-supercategory is *weakly pointed* if every simple object is weakly invertible, and has *split unit* if $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong \mathbb{R}$. We write $\mathcal{W}$ for the class of weakly pointed fusion Π-supercategories with split unit.

Definition 6. Let $\mathrm{Cl}_{p,q}$ be the real Clifford algebra with p positive and q negative squares and its usual $\mathbb{Z}_2$-grading. Fix the central division superalgebras

$$(D_0, \ldots, D_7) = (\mathbb{R}, \mathrm{Cl}_{0,1}, \mathrm{Cl}_{0,2}, \mathrm{Cl}_{0,3}, \mathbb{H}, \mathrm{Cl}_{3,0}, \mathrm{Cl}_{2,0}, \mathrm{Cl}_{1,0}), \tag{9}$$

where $\mathbb{R}$ and $\mathbb{H}$ are concentrated in degree zero. Let $\mathcal{C} \in \mathcal{W}$. An object $X \in \mathcal{C}$ has a *Brauer–Wall type* $r \in \mathbb{Z}_8$ if there is an isomorphism of superalgebras

$$D_r \cong \mathrm{End}_{\mathcal{C}}(X). \tag{10}$$

We call $\mathcal{C}$ a *Brauer–Wall reference category* if every simple object has a unique type, every $r \in \mathbb{Z}_8$ occurs as the type of a simple object, and two simple objects are isomorphic if and only if their types agree.

Theorem 1. *There exists a Brauer–Wall reference category $\mathcal{R} \in \mathcal{W}$.*

Definition 7. A *twist* is a quadruple $t = (G, \rho, \nu, \omega)$, where G is a finite group, $\rho : G \to \mathbb{Z}_8$ is a group homomorphism, and $\nu : G^2 \to \mathbb{Z}_2$, $\omega : G^3 \to \mathbb{R}^\times$ are normalized cochains:

$$\nu(1, g) = \nu(g, 1) = 0, \qquad \omega(1, g, h) = \omega(g, 1, h) = \omega(g, h, 1) = 1. \tag{11}$$

They satisfy

$$\nu(h, k) + \nu(gh, k) + \nu(g, hk) + \nu(g, h) = 0 \tag{12}$$

and

$$\omega(h, k, l)\omega(g, hk, l)\omega(g, h, k) = (-1)^{\nu(g,h)\nu(k,l)}\omega(gh, k, l)\omega(g, h, kl) \tag{13}$$

for all $g, h, k, l \in G$. We write Tw for the class of all twists.

Definition 8. A *gauge* on $s = (G, \rho, \nu, \omega)$ consists of a group isomorphism $\phi : G \to H$ and cochains $\lambda : G \to \mathbb{Z}_2$, $\beta : G^2 \to \mathbb{R}^\times$ with

$$\lambda(1) = 0, \qquad \beta(1, g) = \beta(g, 1) = 1. \tag{14}$$

Its transformed data $(H, \rho', \nu', \omega')$ are given by

$$\rho'(\phi g) = \rho(g), \qquad \nu'(\phi g, \phi h) = \widetilde{\nu}(g, h) := \nu(g, h) + \lambda(g) + \lambda(h) + \lambda(gh), \tag{15}$$

$$\omega'(\phi g, \phi h, \phi k) = \omega(g, h, k)\frac{\beta(gh, k)\beta(g, h)}{\beta(h, k)\beta(g, hk)}(-1)^{\widetilde{\nu}(g,h)\lambda(k) + \lambda(g)\nu(h,k)}. \tag{16}$$

Two twists are *gauge equivalent*, written $s \sim t$, if a gauge transforms the data of s into those of t.

Theorem 2. *The transformed data of a twist under a gauge form a twist.*

Theorem 3. *For every Brauer–Wall reference category $\mathcal{C} \in \mathcal{W}$, there exists a map $\mathcal{P}_{\mathcal{C}} : \mathrm{Tw} \to \mathcal{W}$, called the* twisted pullback*, with the following properties. Every $\mathcal{A} \in \mathcal{W}$ is monoidally superequivalent to $\mathcal{P}_{\mathcal{C}}(t)$ for some $t \in \mathrm{Tw}$, and*

$$\mathcal{P}_{\mathcal{C}}(s) \simeq_{\otimes} \mathcal{P}_{\mathcal{C}}(t) \quad \Longleftrightarrow \quad s \sim t \qquad (s, t \in \mathrm{Tw}). \tag{17}$$

The twist in the reconstruction of $\mathcal{A}$ may be chosen with $\omega(g, h, k) \in \{1, -1\}$ for all g, h, k. For each twist $t = (G, \rho, \nu, \omega)$, the category $\mathcal{P}_{\mathcal{C}}(t)$ has simple objects X_g, indexed by $g \in G$, such that every simple object is isomorphic to some X_g and

$$X_1 = \mathbf{1}, \qquad X_g \cong X_h \Longleftrightarrow g = h. \tag{18}$$

The object X_g has Brauer–Wall type $\rho(g)$. Every simple even retract of $X_g \otimes X_h$ is isomorphic to X_{gh}. Here an even retract of an object Y of a supercategory $\mathcal{C}$ is an object X with morphisms $i : X \to Y$ and $p : Y \to X$ in $\mathcal{C}_0$ such that $p \circ i = \mathrm{id}_X$.

Definition 9. For $\mathcal{C} \in \mathcal{W}$ and a family of objects $(Z_i)_{i \in I}$, the *additive* Π*-hull* $\mathrm{Hull}_{\Pi}(Z_i)$ consists of the objects that are even retracts of finite direct sums of the objects Z_i and $\pi \otimes Z_i$.

Theorem 4. *For $\mathcal{A}, \mathcal{B} \in \mathcal{W}$, there exist $\mathcal{A}\widehat{\boxtimes}\mathcal{B} \in \mathcal{W}$ and a fully faithful monoidal superfunctor*

$$B_{\mathcal{A},\mathcal{B}} : \mathcal{A} \boxtimes \mathcal{B} \longrightarrow \mathcal{A}\widehat{\boxtimes}\mathcal{B} \tag{19}$$

whose image generates the target under finite direct sums, parity shifts, and even retracts:

$$\mathrm{Ob}(\mathcal{A}\widehat{\boxtimes}\mathcal{B}) = \mathrm{Hull}_{\Pi}\big(B_{\mathcal{A},\mathcal{B}}(X, Y)_{X,Y}\big). \tag{20}$$

In the following, we write $\mathcal{P}(t) = \mathcal{P}_{\mathcal{R}}(t)$ and write the chosen simple objects by X_g^t.

Definition 10. Let $\mathrm{Tw}(G)$ be the set of twists with fixed group G. For $s, t \in \mathrm{Tw}(G)$, write $s \sim_G t$ if they are related by a gauge whose group map is id_G. The *stack* of s and t is the full subsupercategory $\mathcal{S}(s, t)$ in $\mathcal{P}(s)\widehat{\boxtimes}\mathcal{P}(t)$ on the objects

$$\mathrm{Hull}_{\Pi}\big(B_{\mathcal{P}(s),\mathcal{P}(t)}(X_g^s, X_g^t)_{g \in G}\big). \tag{21}$$

Theorem 5. *The category $\mathcal{S}(s, t)$ belongs to $\mathcal{W}$ and its inclusion into $\mathcal{P}(s)\widehat{\boxtimes}\mathcal{P}(t)$ is a monoidal superfunctor.*

Definition 11. Define $\epsilon, q : \mathbb{Z}_8 \to \mathbb{Z}_2$ by

$$\epsilon(a) = a \bmod 2, \qquad q(a) = \frac{a(a+1)}{2} \bmod 2, \tag{22}$$

where $a(a+1)/2$ denotes any $b \in \mathbb{Z}_8$ satisfying $2b = a(a+1)$; its reduction modulo 2 is independent of the choice. For $\mathbb{Z}_2$-valued 2-cochains v, z on G, set

$$(v \smile_1 z)(g, h, k) = v(gh, k)z(g, h) + v(g, hk)z(h, k). \tag{23}$$

For $s, t \in \mathrm{Tw}(G)$, write $x = \epsilon \circ \rho_s$, $y = \epsilon \circ \rho_t$, and $c(g, h) = x(g)y(h)$. Their product data $s \star t$ are

$$\begin{aligned} \rho_{s\star t} &= \rho_s + \rho_t, \qquad \nu_{s\star t} = \nu_s + \nu_t + c, \\ \omega_{s\star t}(g, h, k) &= \omega_s(g, h, k)\omega_t(g, h, k)(-1)^{\Xi_{s,t}(g,h,k)}, \end{aligned} \tag{24}$$

where

$$\begin{aligned} \Xi_{s,t}(g, h, k) = (\nu_t \smile_1 \nu_s)(g, h, k) + (c \smile_1 (\nu_s + \nu_t))(g, h, k) \\ + q(\rho_s(g))y(h)y(k) + x(g)x(h)y(h)y(k). \end{aligned} \tag{25}$$

The neutral data are $e_G = (G, 0, 0, 1)$. For $s \in \mathrm{Tw}(G)$, put $v(g,h) = \nu_s(g,h) + x(g)x(h)$. Its inverse data $s^\vee$ are

$$\rho_{s^\vee} = -\rho_s, \qquad \nu_{s^\vee} = v, \qquad \omega_{s^\vee}(g,h,k) = \omega_s(g,h,k)^{-1}(-1)^{\Psi_s(g,h,k)}, \tag{26}$$

where

$$\Psi_s(g,h,k) = (v \smile_1 \nu_s)(g,h,k) + (q(\rho_s(g)) + x(g))x(h)x(k). \tag{27}$$

Theorem 6. *For $s, t \in \mathrm{Tw}(G)$, the data $s \star t$, e_G, and $s^\vee$ are twists on G, and*

$$\mathcal{S}(s,t) \simeq_\otimes \mathcal{P}(s \star t). \tag{28}$$

Theorem 7. *The relation $\sim_G$ is an equivalence relation. The quotient $\mathcal{T}(G) = \mathrm{Tw}(G)/\!\sim_G$ is an abelian group with*

$$[s][t] = [s \star t], \qquad 1 = [e_G], \qquad [s]^{-1} = [s^\vee]. \tag{29}$$

The map

$$d : \mathcal{T}(G) \longrightarrow \mathrm{Hom}(G, \mathbb{Z}_8), \qquad [s] \longmapsto \rho_s \tag{30}$$

is a group homomorphism. Its kernel $\mathrm{SH}_{\mathbb{R}}(G)$ consists of twists with $\rho = 0$ modulo $\sim_G$; on these twists the product is

$$(\nu, \omega) \star (\nu', \omega') = (\nu + \nu',\, \omega\omega'(-1)^{\nu' \smile_1 \nu}). \tag{31}$$

For each finite group G, there is a group isomorphism

$$\Phi_G : \mathcal{T}(G) \xrightarrow{\sim} \mathrm{Hom}(G, \mathbb{Z}_8) \times \mathrm{SH}_{\mathbb{R}}(G) \tag{32}$$

such that

$$\mathrm{pr}_1 \Phi_G = d, \qquad \Phi_G(a) = (0, a) \quad (a \in \mathrm{SH}_{\mathbb{R}}(G)). \tag{33}$$

References

[San24] S. Sanford, *Fusion categories over non-algebraically closed fields*, `arXiv:2401.02354 [math.QA]`.